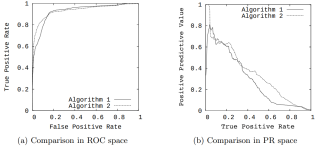
# Introduction to Machine Learning

## Evaluation
## Precision-Recall Curves



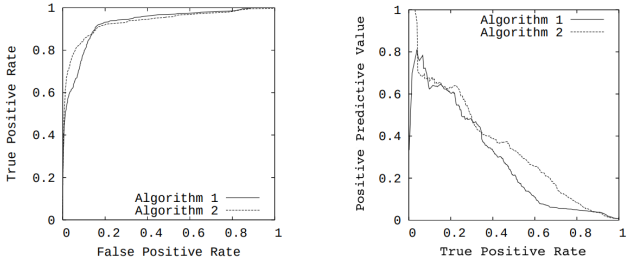

(a) Comparison in ROC space (b) Comparison in PR space

**Learning goals**

- Understand PR curves
- Same as PPV-TPR curve
- Compare to standard TPR-FPR ROC curve

# PRECISION-RECALL CURVES

- Slightly changed ROC plot
- Simply plot precision and recall, instead of TPR-FPR
- Precision = $\rho_{PPV} = \frac{TP}{TP+FP}$, recall = $\rho_{TPR} = \frac{TP}{TP+FN}$
- Might call them TPR-PPV curve
- NB: Both metrics don't depend on TNs

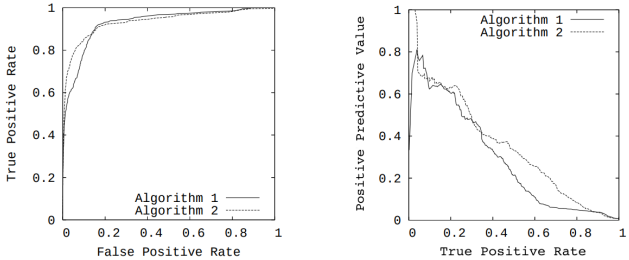



(a) Comparison in ROC space (b) Comparison in PR space

Davis and Goadrich (2006): The Relationship Between Precision-Recall and ROC Curves (URL).

# PRECISION-RECALL CURVES

- Might be better for highly imbal data ($n_- \gg n_+$) than TPR-FPR
- Figure (a): ROC; both learners seem to perform well
- Figure (b): PR; visible room for improvement (top-right=best)
- PR reveals better that algo 2 has advantage over 1

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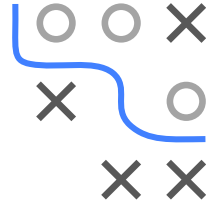
# IMBALANCED DATA



- Assume imbalanced classes with $n_- \gg n_+$
- If neg class large, typically less interested in high TNR = low FPR, but more in PPV
- Large (abs) change in FP yields small change in FPR
- PPV likely more informative

FP=10:

| | True +1 | True -1 |
|---|---|---|
| Pred. Pos | 100 | 10 |
| Pred. Neg | 10 | 9990 |
| Total | 110 | 10000 |

$$\text{TPR} = 10/11$$
$$\text{FPR} = 0.001$$
$$\text{PPV} = 10/11$$

FP=100:

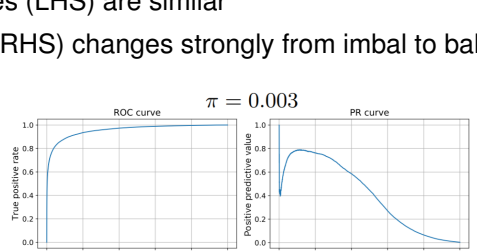
| | True +1 | True -1 |
|---|---|---|
| Pred. +1 | 100 | 100 |
| Pred. -1 | 10 | 9900 |
| Total | 110 | 10000 |

$$\text{TPR} = 10/11$$
$$\text{FPR} = 0.01$$
$$\text{PPV} = 1/2$$

RHS: Given test says +1, it's now a coin flip that this is correct.

# IMBALANCED DATA / 2

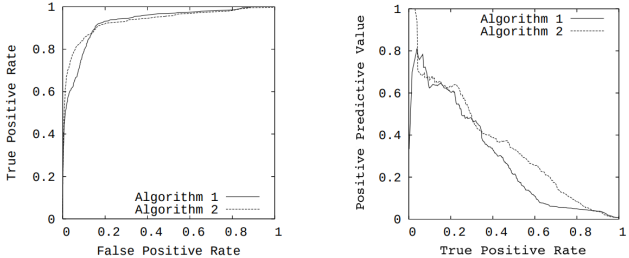
- Top row: Imbal classes with $\pi = 0.003$
- Bottom: balanced with $\pi = 0.5$
- ROC curves (LHS) are similar
- PR curve (RHS) changes strongly from imbal to bal classes




Wissam Siblini et. al. (2004): Master your Metrics with Calibration (URL).

# CONCLUSIONS

- Curve fully dominates in ROC space iff dominates in PR-space
- In imbalanced situations rather use PR than standard TPR-FPR
- If comparing few models on a single task, probably plot both. Then observe and think.
- For tuning: can also use PR-AUC (or partial versions)

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